Jules Van Cauwenberghe

MSc Mechanical Engineering, ETH Zurich | Energy Systems, Risk & Optimization
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Seeking a Semester Project in energy-system modelling, optimization & risk

Profile

MSc student in Mechanical Engineering at ETH Zurich (Energy track), with a BSc from EPFL. Applies system-level risk analysis to railway safety projects under European and Swiss regulations and builds quantitative models (FEA, Python, MATLAB). Currently working with CIMO on the decarbonization of an industrial steam supply.

Education

ETH Zurich (ETHZ) — Zurich, Switzerland
MSc in Mechanical Engineering – Energy Specialization (Tutor: Prof. Russell McKenna) — **Sep 2026 – Present**

- **Autumn 2026:** Modeling & Optimization of Sustainable Energy Systems, Energy Systems Analysis, Stochastic Methods for Engineers, Power Market & Risk Management, Renewable Energy Technologies, Nuclear Energy Conversion, Information Systems for Engineers
- **Planned (spring 2027):** Advanced Techniques for Risk Analysis of Technical Systems, Optimization & ML

École Polytechnique Fédérale de Lausanne (EPFL) — Lausanne, Switzerland
Bachelor of Science in Mechanical Engineering — **Sep 2023 – Jun 2026**

- **Relevant coursework:** Probability & Statistics, Numerical Analysis, Optimization & Operations Research, Control Systems, Foundations of AI, Thermodynamics, Finite Element Method
- **Academic project:** Built an autonomous fruit-harvesting gripper (5-person team), leading embedded control, sensing, UI and actuation (Arduino, OpenCV); ranked 4th out of 51 teams

Experience

Energy Now! (ETH Zurich) – ADAPT Project for CIMO — Zurich, Switzerland
Project Coordinator & Energy System Modeler (team of 5) — **Oct 2026 – Present**

- Assess heat pumps and e-boilers to decarbonize CIMO's 400 GWh/yr steam supply at Monthey, under a reliability constraint of at most 4 supply interruptions per year, each lasting up to 6 h
- Coordinate a team of 5 and CIMO stakeholders; formulate asset-sizing and dispatch optimization models (decision variables, constraints, cost objective) based on clustered typical-week profiles
- Develop a report and 2 models (portfolio sizing, real-time dispatch), in progress and due 14 Dec 2026, for a 2035 scenario with storage and a 130 kt/yr steam gap, on 6 typical weeks at 1-min resolution

Transports Lausannois (TL) — Lausanne, Switzerland
Industry-Sponsored Mechanical Engineering Internship – Railway Safety & Systems Risk — **Feb 2026 – May 2026**

- Identified a passenger-safety gap on the M1 metro line (standing area with no accessible handrail) and initiated the collaboration with TL; led a 2-person team through a system redesign for the 22-unit fleet
- Led system-level risk and interface analysis across 3 rolling-stock generations, ensuring compliance with 5 European (EN) standards (fire and smoke, structural loads, accessibility) and Swiss LHand/OFT regulations
- Developed suspended-handrail concepts in CAD and structural FEA (fatigue safety factors of 4.2–7.8 under normal loads); delivered a dossier preparing industrialization: formal risk analysis, cost estimate, 8 qualified Swiss suppliers

EPFL Xplore – European Rover Challenge (ERC) — Lausanne, Switzerland
Mechanical Design Engineer – Robotic Arm Hardware — **Sep 2025 – Jun 2026**

- Designed the structure of the robotic arm of EPFL Xplore's rover as sole structural engineer; the team won the European Rover Challenge (ERC) 2026 in Kraków, its second consecutive title
- Validated the structure by FEA under system-level load cases (70/35 Nm joint torques, 310 N); folded sheet-metal and welded architecture cut subsystem mass by 750 g (–25%) while maintaining stiffness

Technical Skills

Modelling & Analysis: **Risk analysis**, system-level modelling, **FEA**, numerical methods

Programming: **Python**, MATLAB, C, LabVIEW

Engineering Tools: CATIA, Fusion 360, Inventor, Abaqus

Industrial Context: Railway & energy standards (EN/ISO/OFT/LHand), regulatory compliance, cost estimation

Additional

Academic tutoring – mathematics & physics

Competitive rugby – highest national youth level in France (Crabos), Normandy regional selection

Languages

French Native
English Professional working proficiency
Spanish Intermediate proficiency